

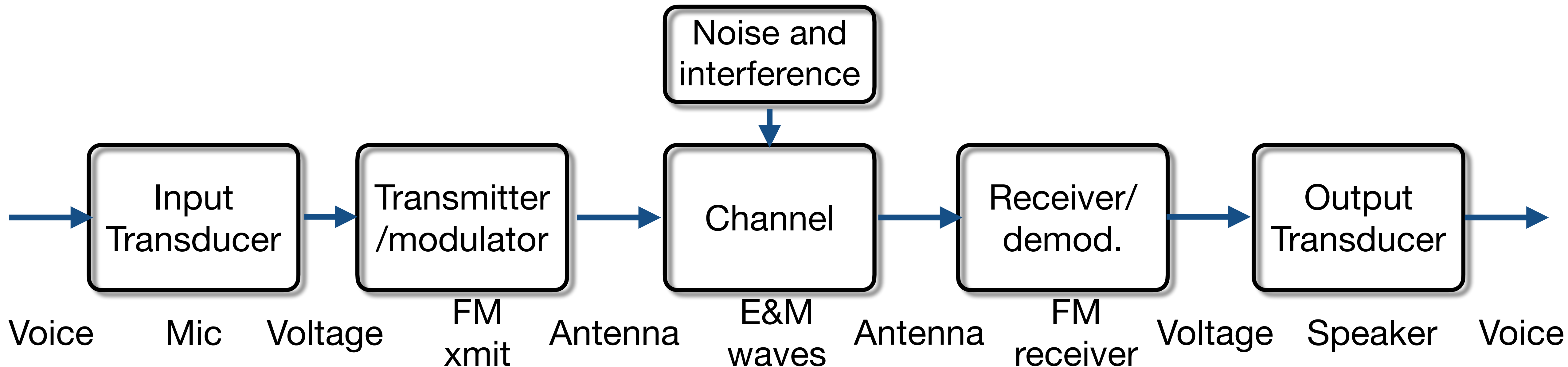
EE121 Midterm 1 Review

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EE121 So Far...

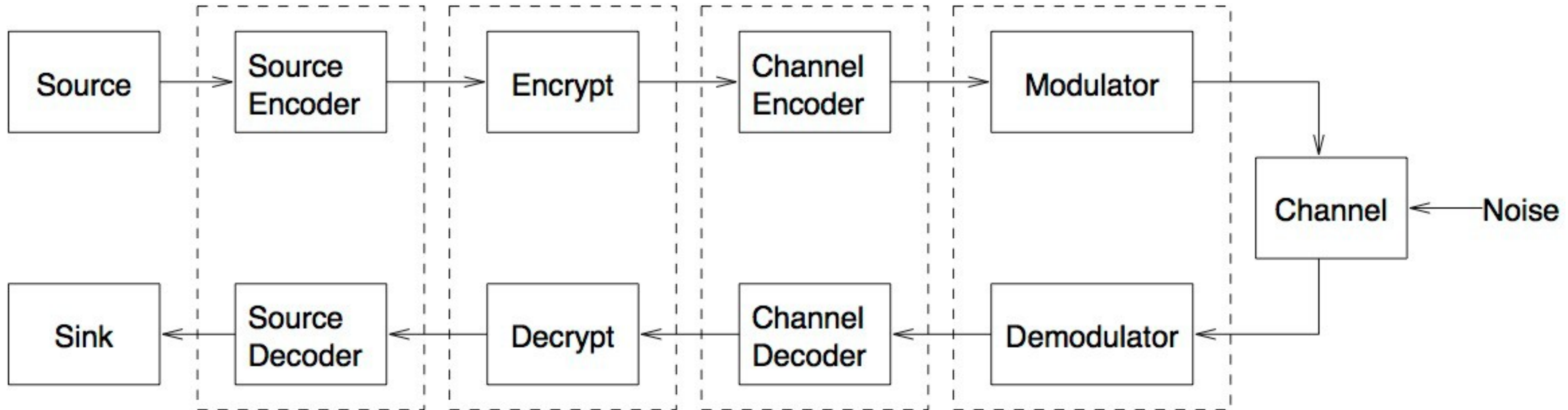
- Intro to communications
- Review of Signals and Systems
- Analog Communications
 - Amplitude Modulation (AM)
 - Angle Modulation (PM/FM)

Analog Communication System



- This is an analog system
- Digital system would transmit bits
 - Inputs/outputs would produce bits, not voltages
 - Transport is still fundamentally analog

Modern Communication System



- Data is compressed, encrypted, and coded for error correction
- Secure link insures privacy
- Guarantees perfect reception (until it doesn't)
- Much more computation (almost free today!)
- Modulator / Channel / Demodulator are all analog

Performance Metrics

Analog

$$\hat{m}(t) \approx m(t)$$

$$\text{MSE} = \frac{1}{T} \int_0^T \left| \hat{m}(t) - m(t) \right|^2 dt$$

$$P_r = \frac{1}{T} \int_0^T \left| m(t) \right|^2 dt$$

$$\text{SNR} = \frac{P_r}{\text{MSE}} = \frac{P_r}{\sigma^2}$$

Digital

Data rate R in bits/s

$$P_e = P \left\{ \hat{b} \neq b \right\}$$

$$C = B \log_2(1 + \text{SNR})$$

B : Channel Bandwidth (Hz)

C : bits/s

Signals and Systems

- Analog and Digital Signals
- Useful Functions (rect, sinc, tri, sinusoids, complex exponentials, delta function)
- Energy and Power
- LTI Systems (Convolution)
- Fourier Transforms and Properties
- Baseband and passband (complex baseband (I and Q))
- Parseval, Energy Density Spectrum, Autocorrelation
- Filters

Consider the following two passband signals:

$$u_p(t) = \text{sinc}(2t)\cos(100\pi t)$$

$$v_p(t) = \text{sinc}(t)\sin(101\pi t + \pi/4)$$

a) Find the complex envelopes $u(t)$ and $v(t)$ for u_p and v_p , respectively, with respect to the frequency reference $f = 50$ Hz.

Consider the following two passband signals:

$$u_p(t) = \text{sinc}(2t)\cos(100\pi t)$$

$$v_p(t) = \text{sinc}(t)\sin(101\pi t + \pi/4)$$

b) What's the bandwidth of $u_p(t)$? What's the bandwidth of $v_p(t)$?

Consider the following two passband signals:

$$u_p(t) = \text{sinc}(2t)\cos(100\pi t)$$

$$v_p(t) = \text{sinc}(t)\sin(101\pi t + \pi/4)$$

c) Find the inner product $\langle u_p, v_p \rangle$, using the result from a).

Consider the following two passband signals:

$$u_p(t) = \text{sinc}(2t)\cos(100\pi t)$$

$$v_p(t) = \text{sinc}(t)\sin(101\pi t + \pi/4)$$

d) Find the convolution $y_p(t) = (u_p * v_p)(t)$, using the result from a).

Design following Windowed sinc Filters:

a) 50 kHz and a transition of 5 kHz

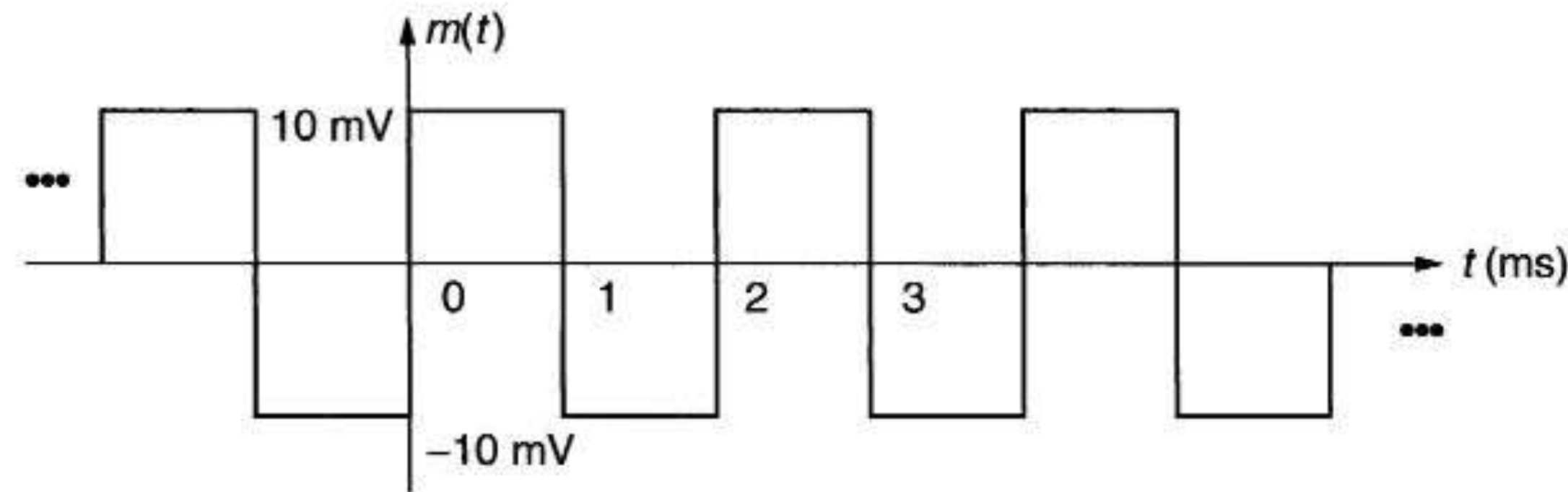
b) 1 MHz and a transition of 100 kHz

c) Is it possible to adapt these filters into bandpass filters?

Amplitude Modulation (AM)

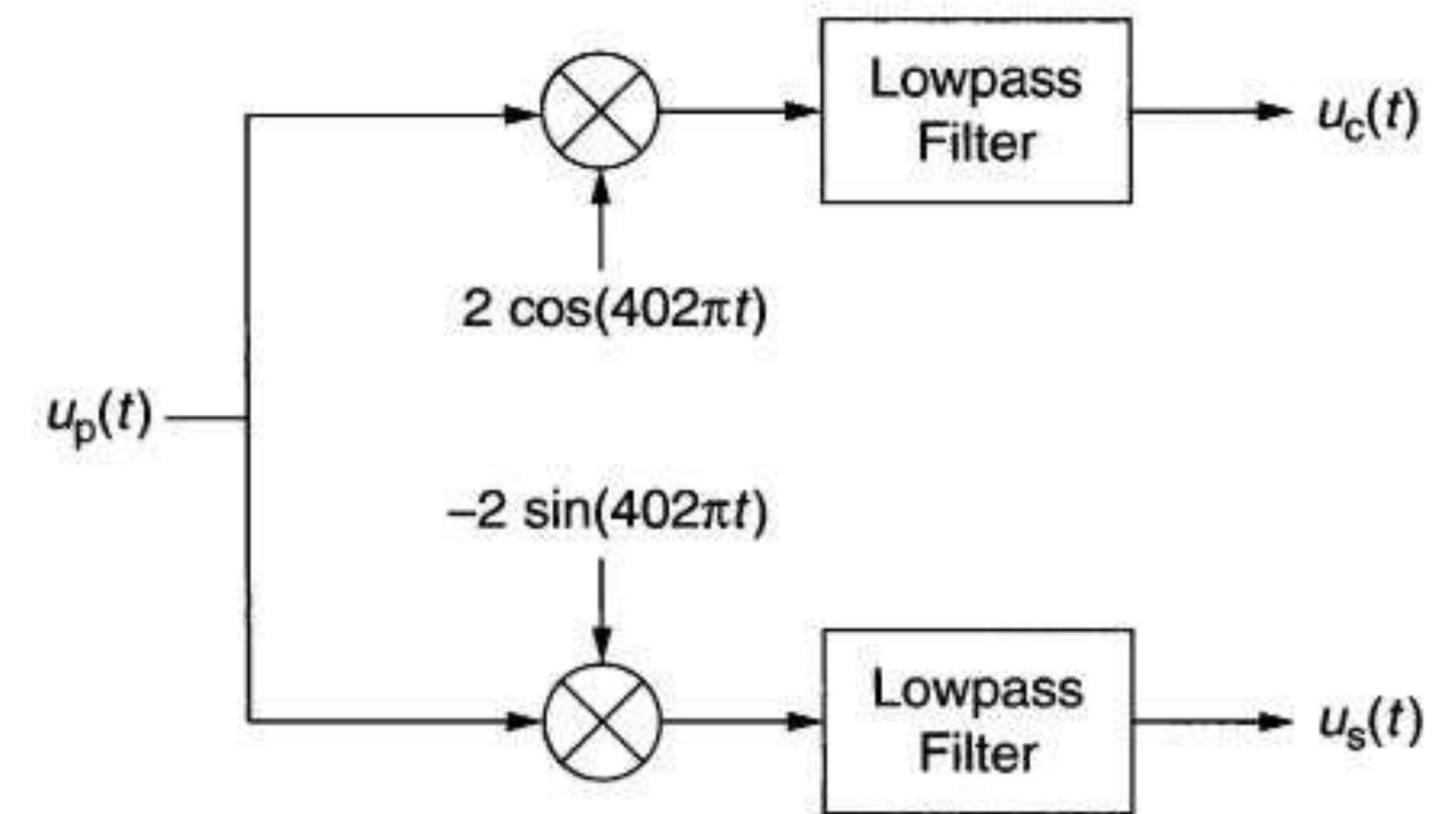
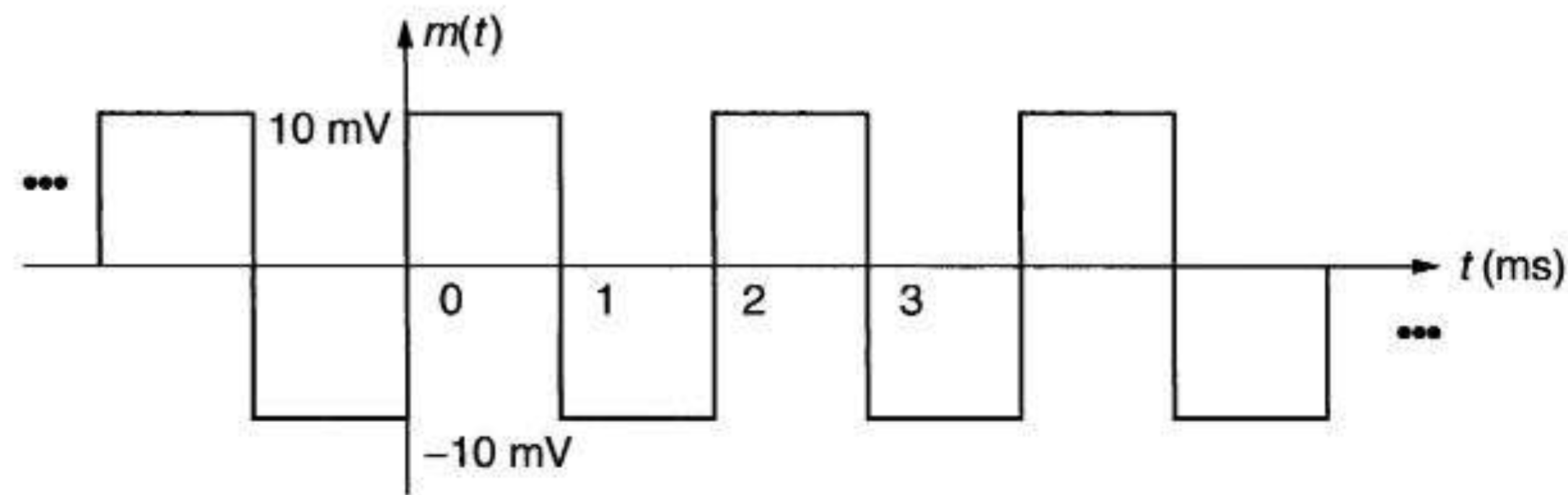
- DSB-SC
- Synchronous receiver
- Standard broadcast AM
- Envelope Detection
- Power efficiency of AM
- Single Sideband (SSB)
- QAM

Consider the AM signal $u_p(t) = (A + m(t))\cos(400\pi t)$ (t in ms) with message signal $m(t)$ shown below with $A = 10$ mV.



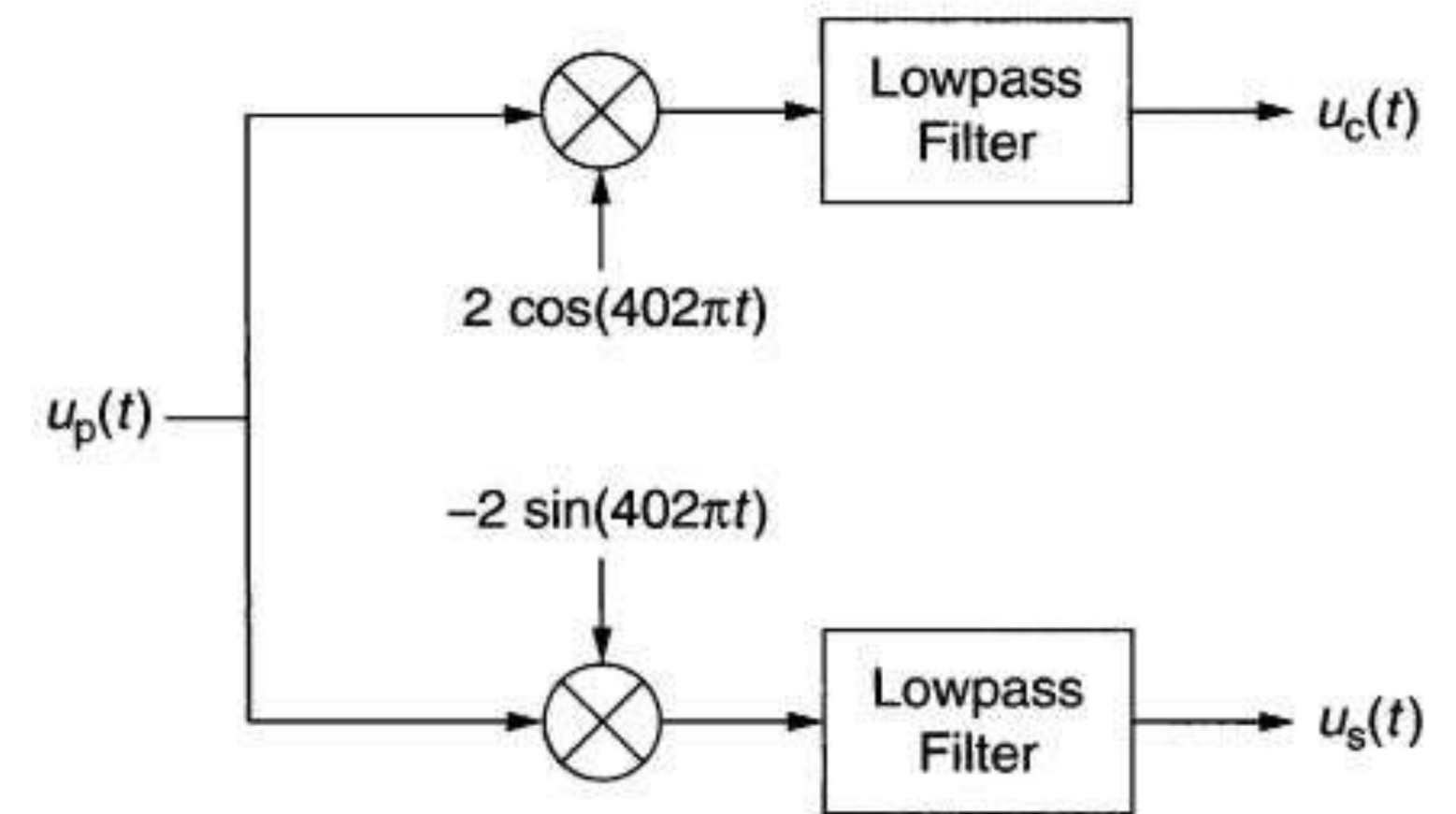
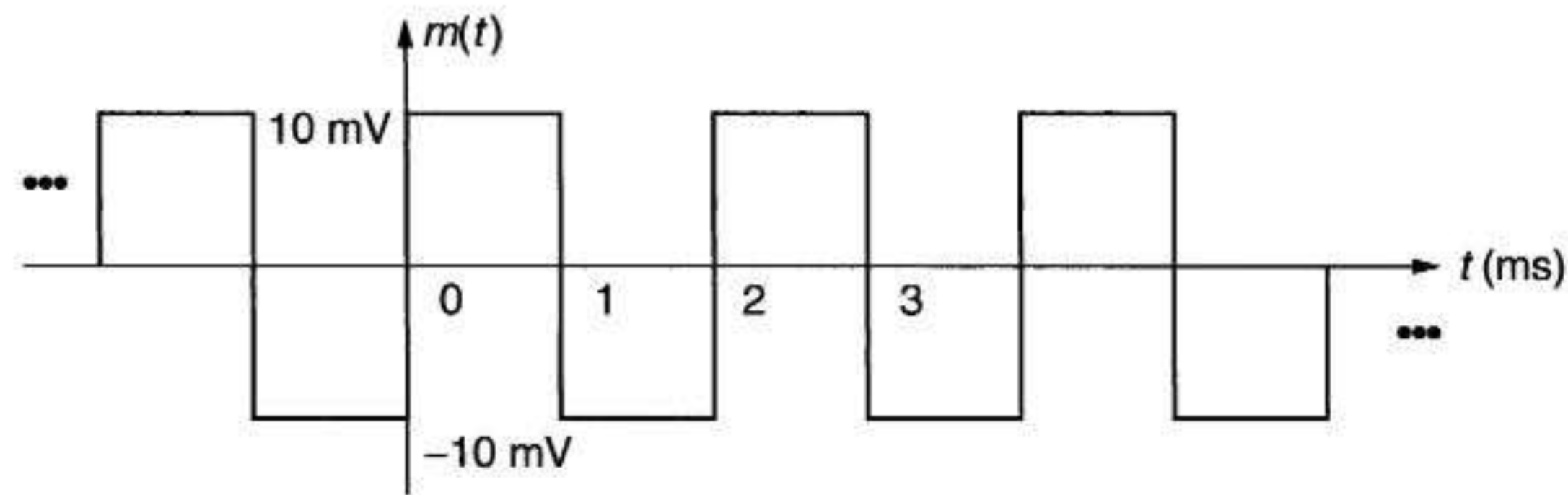
a) If the AM signal is demodulated using an envelope detector with an RC filter, how should you choose C if $R = 500\Omega$? Try to ensure that the first harmonic (i.e., the fundamental) and the third harmonic of the message are reproduced with minimal distortion.

Consider the AM signal $u_p(t) = (A + m(t))\cos(400\pi t)$ (t in ms) with message signal $m(t)$ shown below with $A = 10$ mV.



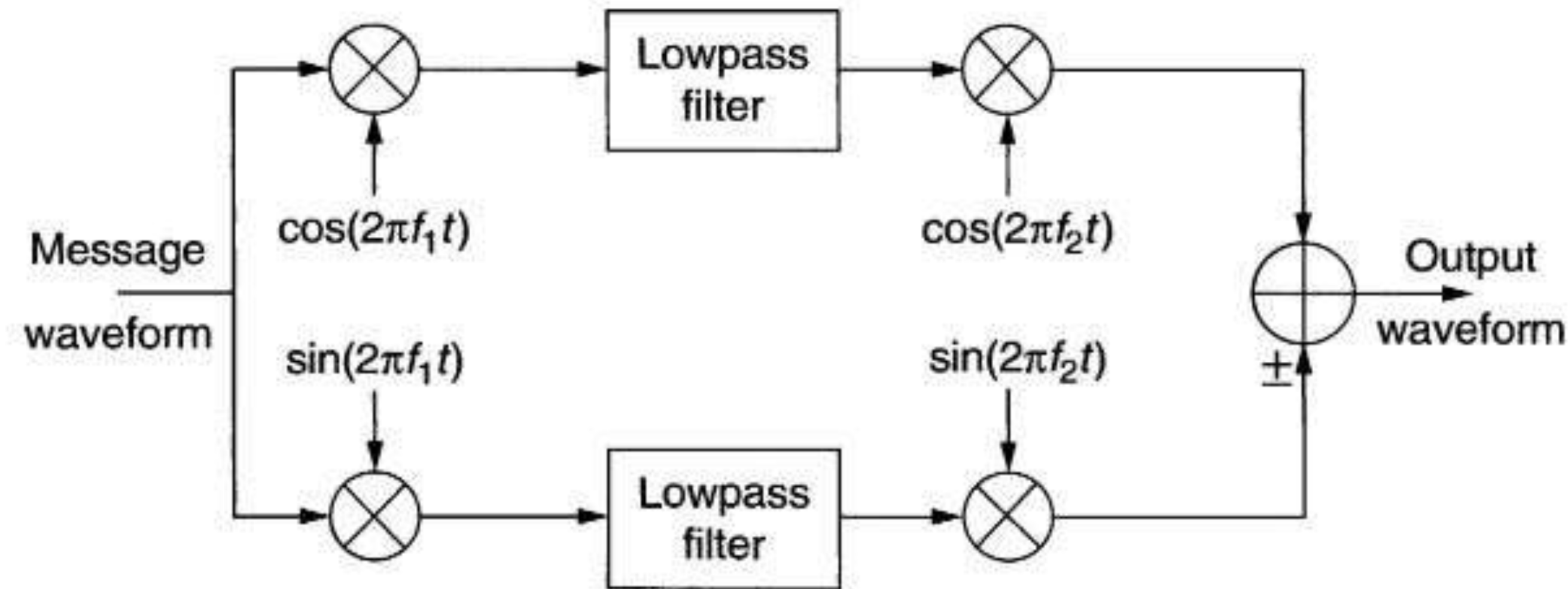
b) Now, consider an attempt at synchronous demodulation, where the AM signal is down-converted using a 201-kHz LO, as shown above. Find and sketch the I and Q components, $u_c(t)$ and $u_s(t)$, for $0 \leq t \leq 2$ (t in ms).

Consider the AM signal $u_p(t) = (A + m(t))\cos(400\pi t)$ (t in ms) with message signal $m(t)$ shown below with $A = 10$ mV.

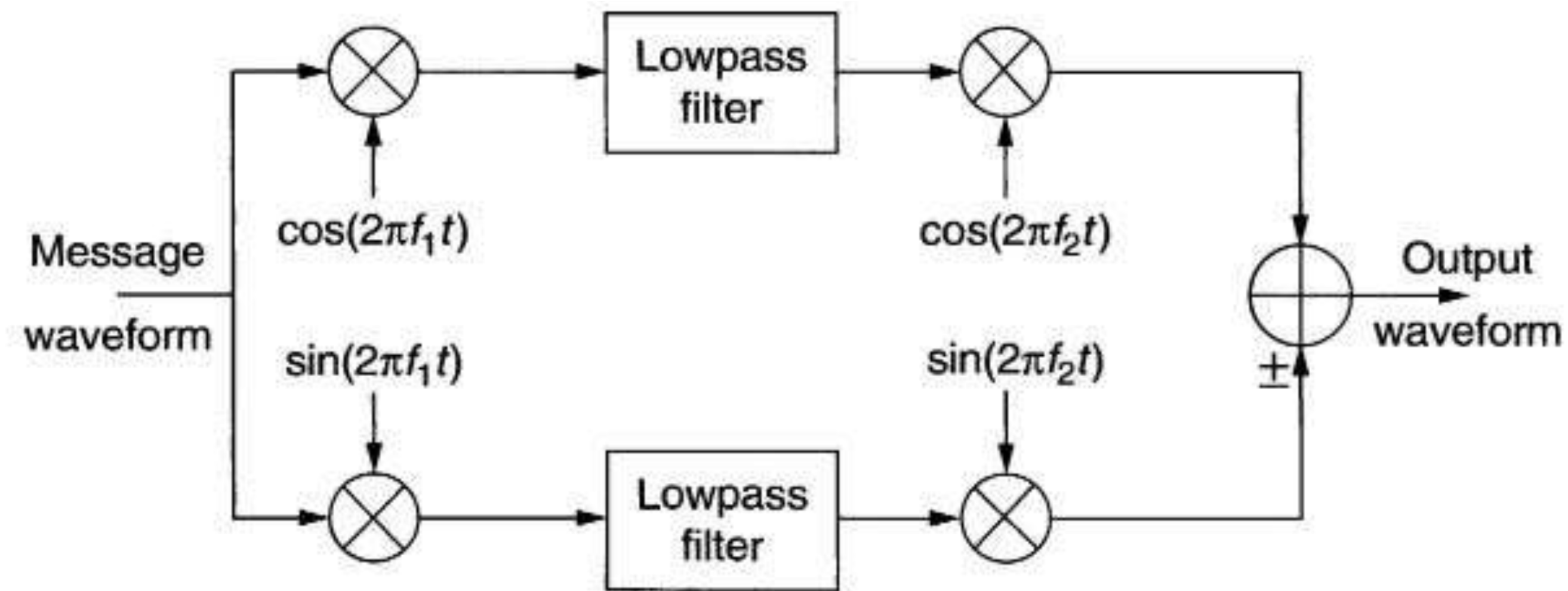


c) Describe how you would recover the original message $m(t)$ from the downconverter outputs $u_c(t)$ and $u_s(t)$, drawing block diagrams as needed.

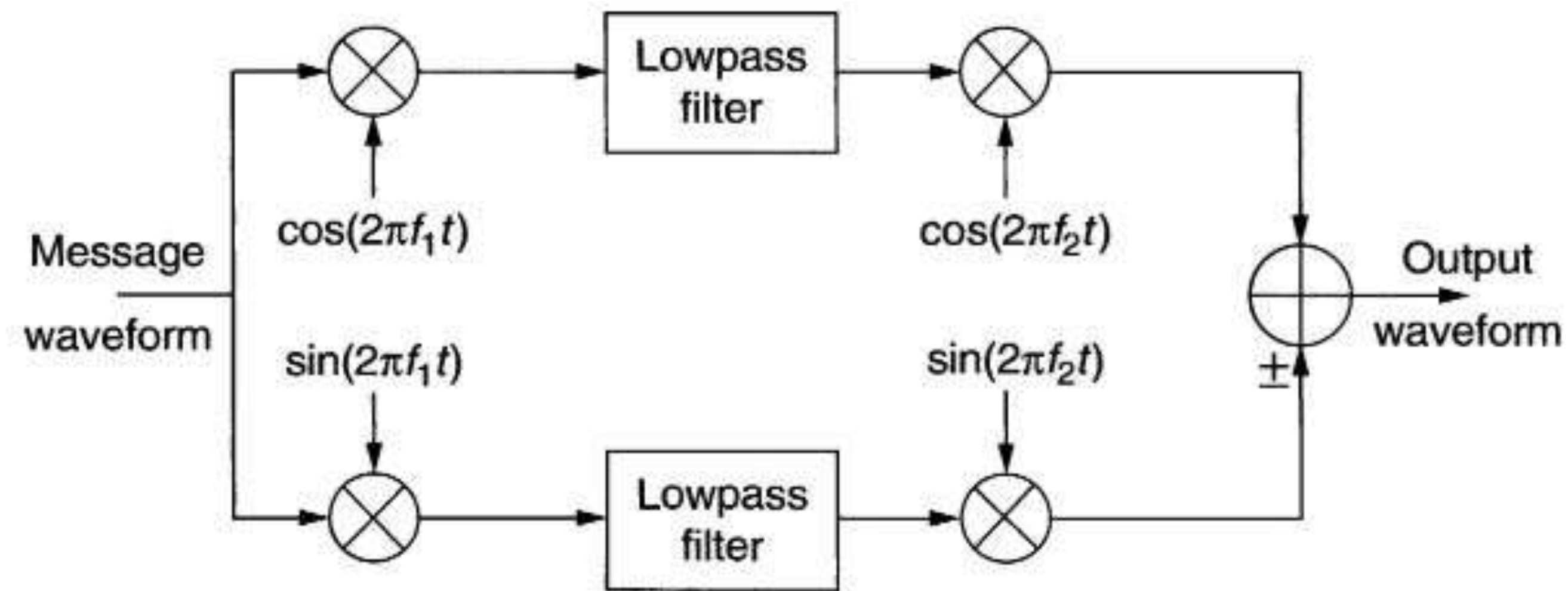
Below is a block diagram of *Weaver's* SSB modulator, which works if we choose f_1 , f_2 , and the bandwidth of the lowpass filter (LPF) appropriately. Let us work through these choices for a waveform of the form $m(t) = A_L \cos(2\pi f_L + \phi_L) + A_H \cos(2\pi f_H + \phi_H)$, where $f_H > f_L$ (the design choices we obtain will work for any message whose spectrum lies in the band $[f_L, f_H]$).



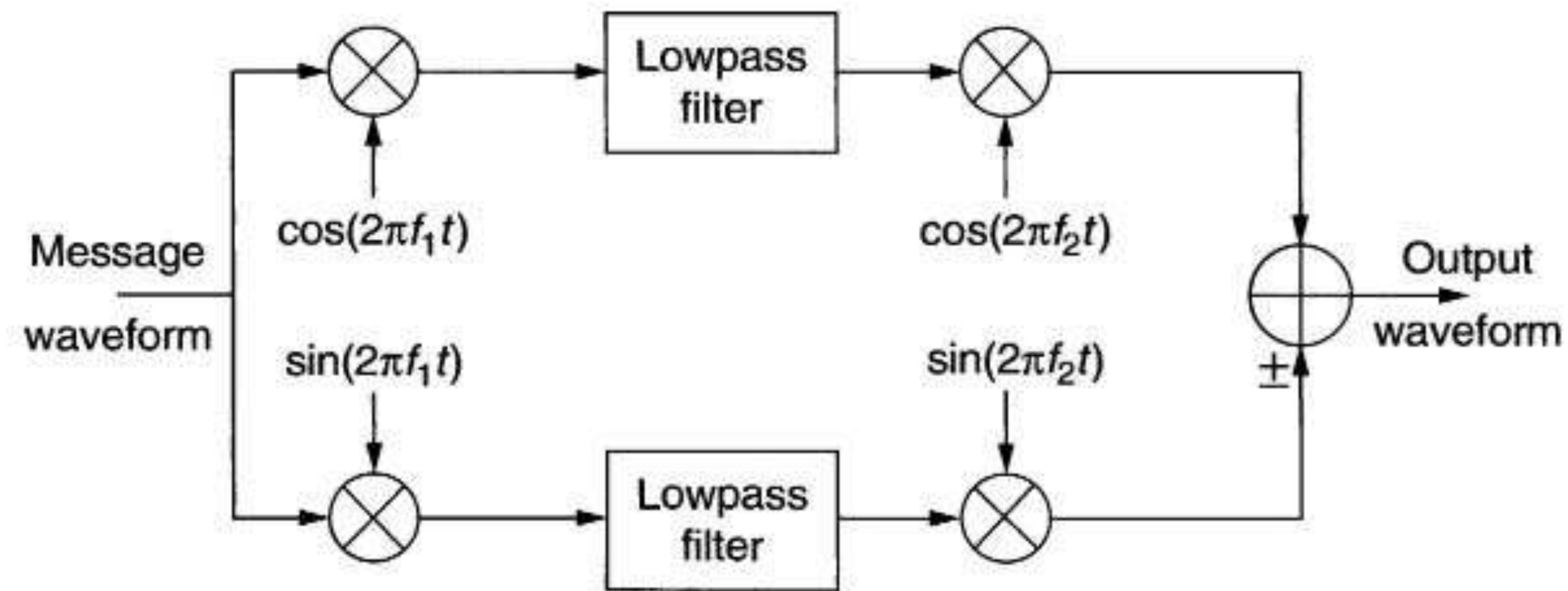
a) For $f_1 = (f_L + f_H)/2$ (i.e., choosing the first LO frequency to be in the middle of the message band), find the time-domain waveforms at the outputs of the upper and lower branches after the first mixer.



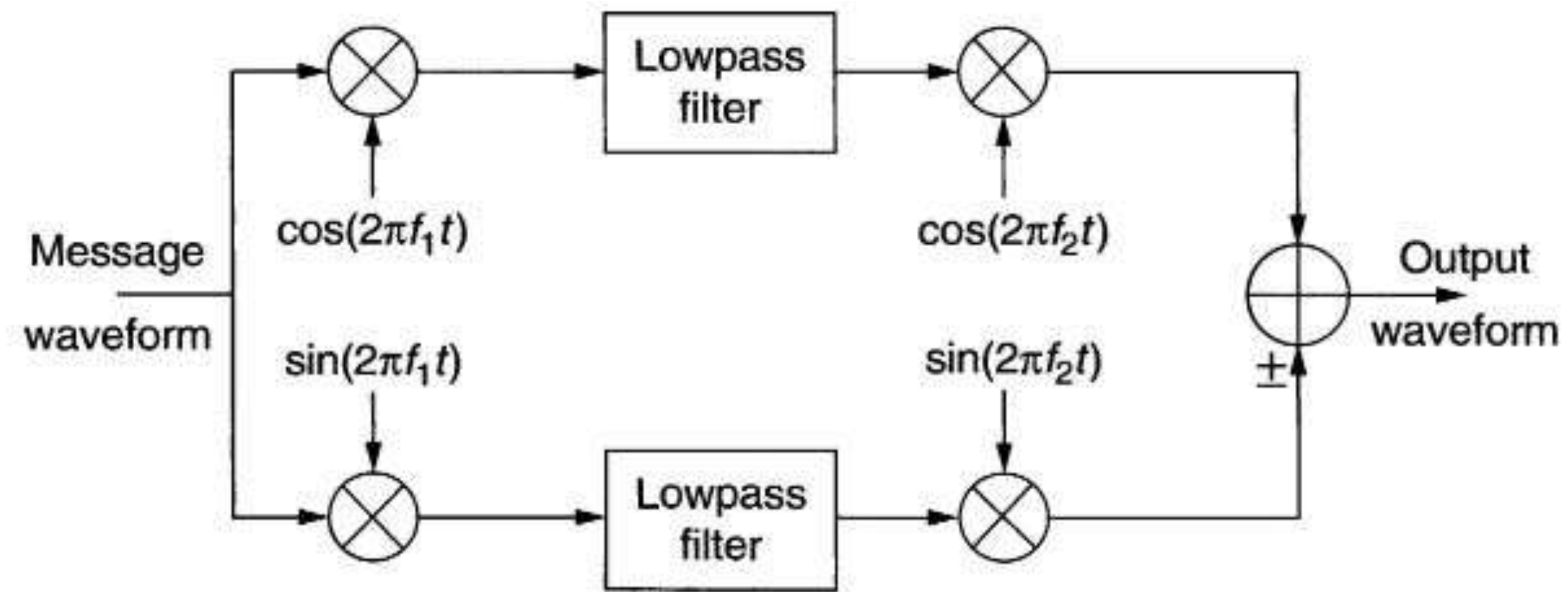
a) For $f_1 = (f_L + f_H)/2$ (i.e., choosing the first LO frequency to be in the middle of the message band), find the time-domain waveforms at the outputs of the upper and lower branches after the first mixer.



b) Choose the bandwidth of the LPF to be $W = (f_H + 2f_L)/2$ (assume that the LPF is ideal). Find the time-domain waveforms at the outputs of the upper and lower branches after the LPF.



c) Now, assuming that $f_2 \gg f_H$, find a time-domain expression for the output waveform, assuming that the upper and lower branches are added together. Is this an LSB or USB waveform? What is the carrier frequency?

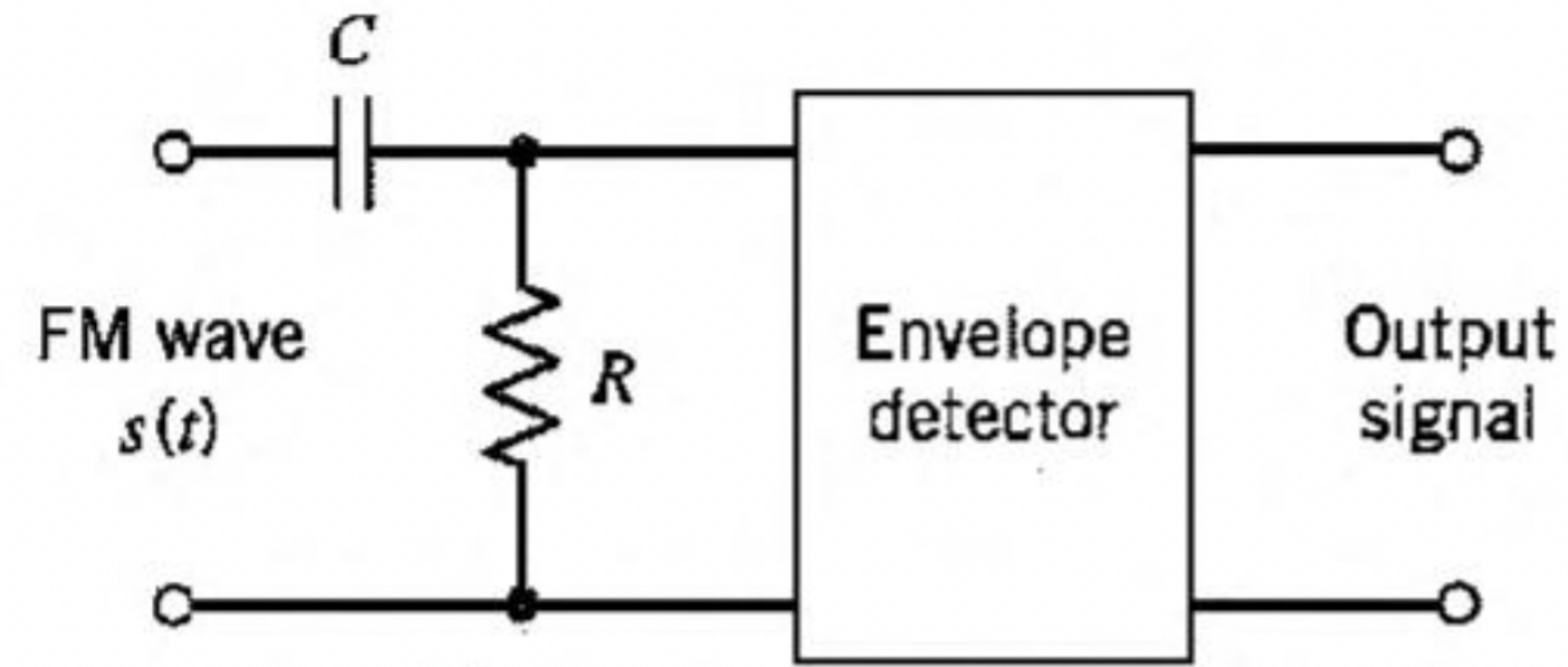


d) Repeat c) when the lower branch is subtracted from the upper branch.

Angle Modulation

- Frequency and Phase Modulation (FM/PM)
- FM Modulation and demodulation
- Differentiators
- FM Pre-emphasis
- Broadcast FM
- Brief PLLs

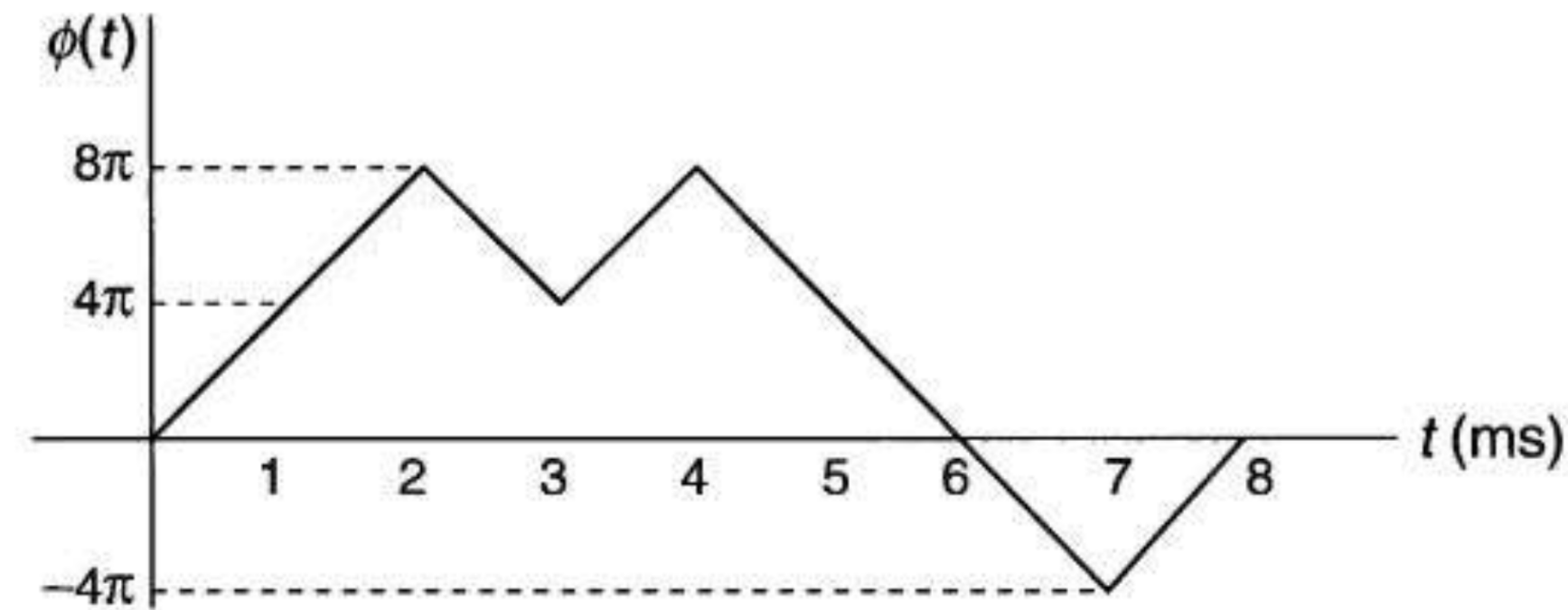
Determine the output of the following frequency demodulator.



$$s(t) = A_c \cos \left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau) d\tau \right)$$

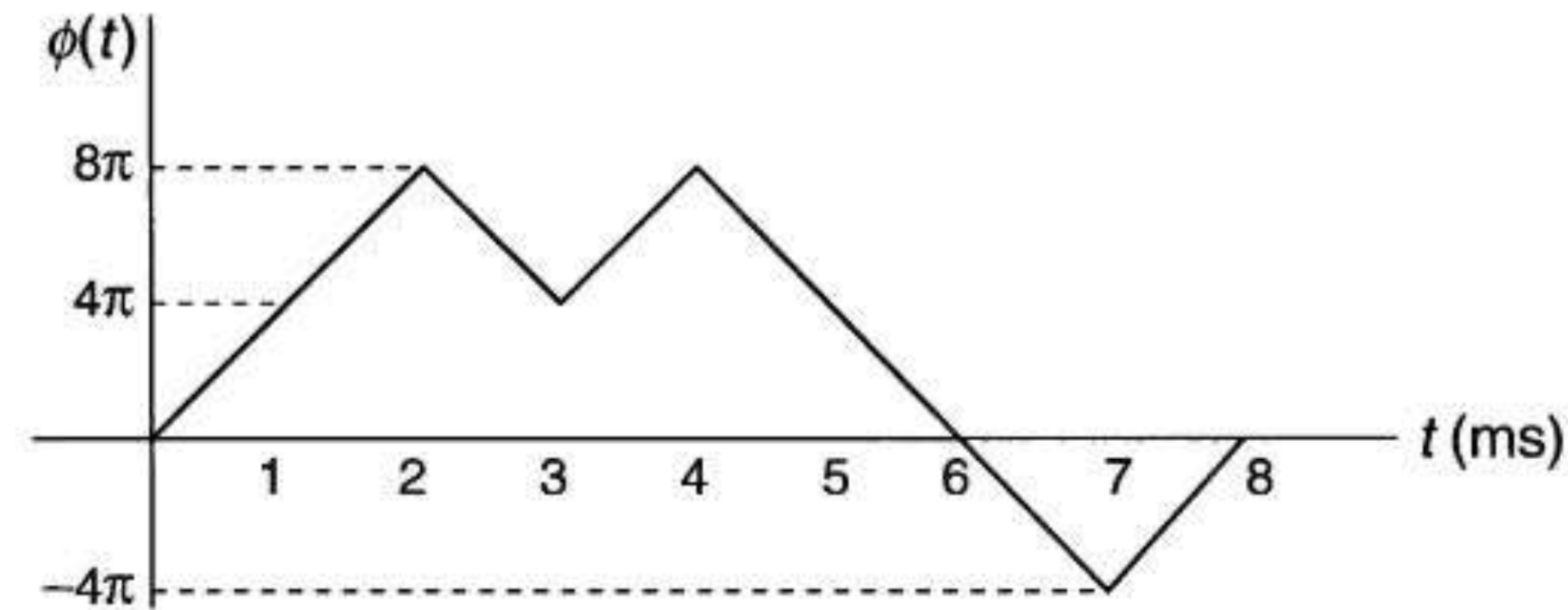
Assume that $2\pi f_c RC \ll 1$.

The output of an FM modulator is the bandpass signal $y(t) = 10 \cos(300\pi t + \phi(t))$, where the unit of time is milliseconds, and the $\phi(t)$ is shown below.



a) Supposing that $y(t)$ is the output of a VCO with frequency deviation 1 kHz/mV and quiescent frequency 149 kHz, find and sketch the input to the VCO.

The output of an FM modulator is the bandpass signal $y(t) = 10 \cos(300\pi t + \phi(t))$, where the unit of time is milliseconds, and the $\phi(t)$ is shown below.



b) Use Carson's formula to estimate the bandwidth of $y(t)$, clearly stating the approximations that you make.